Structured Summarization and Evaluation of Real-World Work Chats

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Abstract

This work studies structured summarization of real-world work chat conversations. Summaries are generated in a task-oriented format separating context, decisions, action items, and open questions. A dataset of anonymized chat fragments is constructed, with a limited gold subset created via an LLM-assisted annotation pipeline based on GPT-4.1 and human validation. Multiple large language models are evaluated using component-wise automatic metrics and judge-based scoring. The results reveal observable differences in generation behavior across evaluated models, highlighting trade-offs between precision, completeness, and faithfulness in structured chat summarization. Project code: https://github.com/va-av-8/work-chat-summarization-eval/tree/main

1 Introduction

Instant messaging platforms have become a primary medium for work-related communication. Decisions, task assignments, and clarifications are often discussed in short chat messages rather than in formal documents. While this enables fast collaboration, important outcomes of discussions are easily lost in chat histories, making it difficult to track responsibilities and unresolved issues.

Automatic summarization of work chats is therefore an important natural language processing task. However, real-world work chats are noisy, multi-speaker, and highly informal. They contain short acknowledgements, off-topic messages, and implicit references to tasks or decisions, which makes unstructured summaries and simple extractive approaches insufficient for practical use.

Most existing work on conversational summarization focuses either on spoken meeting transcripts or on synthetic chat datasets, and typically produces unstructured free-text summaries. At the same time, research on action item and decision extraction treats these problems separately from summary generation. As a result, current approaches do not fully address the requirements of task-oriented work communication.

This work focuses on structured summarization of real-world work chats. Short fragments of internal chat conversations are summarized into a fixed schema that explicitly separates conversational context, decisions, action items, and open questions. Such a structure is intended to better support practical use cases, where summaries are expected to highlight actionable information.

Due to the high cost of manual annotation, an LLM-assisted annotation process followed by manual validation is used to construct a limited gold evaluation subset. Evaluation is performed using a combination of component-wise automatic metrics and large language model–based judgments, allowing both factual accuracy and perceived overall usefulness of structured summaries to be assessed.

The goal of this study is to investigate the generation and evaluation of structured summaries for real-world work chats and to provide a reproducible evaluation framework tailored to this task.

1.1 Team

This project was completed by a single author.

2 Related Work

Automatic summarization of conversational data has been widely studied in the context of meeting transcripts and asynchronous communication. Early work on meeting summarization focused on long spoken interactions and aimed to capture the overall discussion flow rather than task-oriented outcomes. For example, Murray et al. proposed extractive and abstractive approaches for meeting summarization on the AMI corpus [Murray et al., 2005], while Galley et al. investigated discourse-based methods for segmenting and summarizing multi-party conversations [Galley et al., 2006].

Summarization of email conversations represents another related line of research. Zhong et al. studied extractive summarization of email threads, selecting representative sentences to summarize discussion chains [Zhong et al., 2002]. Although this work addressed work-related textual communication, email conversations differ substantially from modern work chats in interaction dynamics, message length, and noise patterns.

Neural chat summarization has gained popularity with the introduction of the SAMSum dataset [Gliwa et al., 2019], which contains short messenger-style dialogues paired with abstractive summaries. However, SAMSum dialogues are synthetically generated and lack properties typical for real work chats, such as task coordination, implicit decision-making, and responsibility assignment. In addition, summaries in SAMSum are unstructured free-text paragraphs and do not explicitly model action items or unresolved questions.

A related research direction focuses on extracting task-related information from conversations. Purver et al. studied task-based summarization of meetings, aiming to identify decisions and commitments [Purver et al., 2007], while Chen et al. formulated action item extraction as a supervised learning problem applied to meeting transcripts [Chen et al., 2018]. These approaches explicitly target actionable content but typically treat extraction separately from summary generation and rely on relatively small annotated subsets due to the high cost of manual labeling.

Evaluation of abstractive summarization remains challenging. Traditional automatic metrics such as ROUGE [Lin, 2004] have limited correlation with human judgments for conversational and highly abstractive summaries. Recent work proposes using large language models as evaluators. Liu et al. introduced G-EVAL, demonstrating that LLM-based evaluation can approximate human judgments on small, manually validated evaluation sets [Liu et al., 2023]. Similarly, Chiang and Lee showed that large language models can serve as effective judges in pairwise comparison settings [Chiang and Lee, 2023].

In contrast to prior work, this study focuses on real-world work chats and targets structured summaries that explicitly separate conversational context, decisions, action items, and open questions. Given the annotation complexity, we adopt an LLM-assisted annotation setup followed by manual validation to construct a limited gold subset, which is consistent with practices in task-oriented dialogue summarization and LLM-based evaluation research [Purver et al., 2007; Liu et al., 2023]. For evaluation, we combine component-wise automatic metrics with LLM-based judgments to address limitations of both approaches.

3 Model Description

This section describes the proposed approach for generating structured summaries of work chat conversations. The focus is on the formulation of the generation task, the output representation, and the choice of language models used to produce candidate summaries.

3.1 Task Formulation

The task is formulated as structured abstractive summarization of short multi-speaker chat conversations. Given a dialogue consisting of a sequence of messages exchanged by several participants, the goal is to generate a summary that explicitly separates different types of information relevant for work coordination.

Let a dialogue be represented as

$$D = \{m_1, m_2, \ldots, m_n\},$$

where each message m_i contains a timestamp, a speaker identifier, and a text segment.

The output is a structured summary

$$S=\{C, D, A, Q\},$$

where:

- C is a short textual description of the conversational context,
- D is a list of decisions explicitly made in the dialogue,
- A is a list of action items, optionally associated with assignees and deadlines,
- Q is a list of open questions that remain unresolved.

This formulation differs from traditional free-text summarization by enforcing an explicit structure that aligns with practical usage scenarios.

3.2 Structured Output Representation

All summaries are generated in a fixed JSON format with predefined fields corresponding to the components described above. This representation constrains the generation process and reduces ambiguity in model outputs. It also enables direct comparison of individual components across different models.

Each action item is represented as a tuple consisting of a textual description and optional metadata fields (assignee and deadline). These fields are not required to be present in every case, reflecting the implicit nature of task assignments in informal chat communication.

No post-generation rewriting or aggregation is applied to the model outputs. Each summary is generated in a single pass based solely on the dialogue input and the instruction prompt.

Here is the summary JSON schema:

```
{
  "Context": "1-3 sentences about the essence of the discussion.",
  "Decisions": ["Brief formulations of explicit solutions."],
  "Actions": [
    { "assignee": "Who or null", "description": "What to do or null", "deadline": "When or  null" }
  ],
  "Questions": ["Open questions without answers in dialogue."]
}
```

3.3 Prompt-based Generation

All candidate summaries are generated using instruction-following large language models. Each model receives the same task description and output schema as part of the prompt, ensuring comparability across different systems. The dialogue is provided as a linearized text sequence that preserves message order and speaker attribution.

The approach relies exclusively on prompt-based inference and does not involve any model fine-tuning or additional training. This design choice allows the method to be applied uniformly to models with different architectures and deployment settings.

3.4 Model Selection

A limited set of language models is selected to generate candidate structured summaries. The purpose of this selection is not to provide an exhaustive benchmark of available systems, but to compare a small number of representative models with different capacities, costs, and deployment settings under a unified generation setup.

Three locally deployed open-source models are included, covering lightweight and medium-capacity local deployment scenarios:

Qwen2.5-3B-Instruct represents a compact model suitable for low-resource or privacy-sensitive environments, where local deployment and computational efficiency are critical.

LLaMA-3.1-8B-Instruct represents a higher-capacity locally deployed model with stronger reasoning and generation capabilities, serving as a medium-scale local baseline.

Mistral-7B-Instruct (local) is included as an additional medium-capacity local model from the Mistral family. Its inclusion enables a within-family comparison between local deployment and API-based deployment, complementing cross-family comparisons among open-source models.

In addition to local models, two API-based models are used:

GPT-4o represents a high-capacity general-purpose language model with strong instruction-following abilities and robust structured output generation.

Mistral Large represents a competitive non-GPT API-based model. Together with the local Mistral-7B baseline, it enables analysis of how deployment constraints and model scaling within the same model family affect structured summarization quality.

This selection enables comparison across multiple dimensions, including model size, deployment constraints, and generation quality, while keeping the experimental setup tractable. Restricting the number of evaluated models is a deliberate design choice, motivated by the cost of inference and the need for careful qualitative and quantitative evaluation of structured outputs. This approach allows for detailed analysis without compromising reproducibility or clarity of results.

3.5 Method Summary

In summary, the proposed method formulates work chat summarization as a structured generation task and applies prompt-based inference with a fixed output schema. Multiple language models are used to generate candidate summaries, enabling analysis of how model capacity and deployment setting affect the quality of structured outputs.

4 Dataset

This section describes the dataset used for structured summarization of work chat conversations.

4.1 Data Source and Collection

The dataset consists of anonymized fragments of real-world internal work chats collected from instant messaging platforms. The data was obtained from personal work communication archives, with all personal identifiers removed to ensure privacy. The dataset is intended solely for research purposes.

Chat logs are segmented into short conversation fragments, each containing 20 consecutive messages. This batch size was chosen to reflect realistic discussion units while keeping input length manageable for large language models.

4.2 Preprocessing and Anonymization

All chat data undergoes normalization and anonymization. Speaker names, usernames, and chat titles are replaced with anonymous identifiers. Message texts are normalized by flattening formatting artifacts and removing empty messages. Hyperlinks are preserved in textual form, non-informative messages containing only emojis or acknowledgements are deleted.

Each chat fragment is converted into a linearized dialogue representation preserving message order, timestamps, and anonymized speaker identifiers.

```
[30/30] dialogue_id: Chat_003__batch_00004

2025-10-29T10:44:06 | Speaker_B: org – organization, loc – location и т.д.
2025-10-29T10:44:45 | Speaker_A: а, понял
2025-10-29T10:44:54 | Speaker_B: в целом на этом этапе нам это не так важно – какая сущность зафиксирована, это скорее для отладки. Факт детекции хоть какой-то сущности у
нас планировался как пропуск для следующего этапа – extraction с участием LLM. Работает быстро, у меня локально на CPU задержка на ответ в ТГ секунда-две
2025-10-29T10:46:14 | Speaker_A: а поле is-task-related ты сам задал или оно из коробки как-то работает?
2025-10-29T10:47:43 | Speaker_B: нет, не из коробки. пока вот такой костыль has_person = any(e["label"] == "PER" for e in detected_entities) has_date = any(e["label"] ==
"DATE" for e in detected_entities) is_task_related = has_person or has_date
2025-10-29T10:47:44 | Speaker_A: короче нам репозиторий нужен общий жееестьб 😁
2025-10-29T10:49:15 | Speaker_B: мы можем <user_5> подождать с репой, или могу пока скинуть что есть, если сейчас надо
2025-10-29T10:59:27 | Speaker_D: всм?
2025-10-29T12:01:58 | Speaker_B: вроде обсуждали, что ты сделаешь репозиторий от нити, которым мы будем пользоваться
2025-10-29T12:02:51 | Speaker_B: ...и сервак (может путаю)
2025-10-29T12:03:15 | Speaker_D: просто пустой репо?
2025-10-29T12:03:20 | Speaker_D: или вы хотите бота?
2025-10-29T12:04:12 | Speaker_B: можно пустой, можно с тем, что уже было раньше – но неуверен, что требуется
2025-10-29T13:12:29 | Speaker_D: сегодня сделаю!
2025-10-29T23:42:23 | Speaker_E: Начала экспериментировать с саммари. В целом, есть из чего выбрать по моделям, которые даже без дообучения хорошо себя показывают на задачах
саммаризации диалогов. Вопрос, насколько для нас критично latency и к какому мы стремимся? По идее, пользователь не ждёт мгновенной генерации задачи, качество генерации
важнее? Пока экспериментирую в коллабе, закладываюсь на то, что вместе с сообщениями приходят выделенные фильтром сущности, но пока экспериментирую со значениями None в них.
Также, нашла какие-то примерно подходящие датасеты по саммаризации рабочих встреч и переписок, но подробно ещё не смотрела их ноутбук: <url>
2025-11-02T10:40:42 | Speaker_A: Ребята, всем привет! Мы можем сегодня сместить встречу на 16:30?
2025-11-02T11:08:21 | Speaker_B: Я тоже за! Непредвиденные обстоятельства
2025-11-02T11:58:38 | Speaker_E: Привет! Я за. Было бы здорово, если бы о переносах получалось узнавать чуть заранее, иногда уже есть другие планы. В следующее вс точно не
смогу в 16:30. Ещё, думаю, может, попробуем перенести наши встречи на будни? Например, на четверг в 19.00/19:30? Сейчас в сб у меня пары, в вс встреча, получается совсем без
выходных, становится тяжеловато)
2025-11-02T11:59:08 | Speaker_D: Я сегодня не смогу
2025-11-02T11:59:17 | Speaker_D: а кроме четверга?
```

Figure 1: Example of a chat fragment.

4.3 Dataset Split

The dataset is divided into two subsets with distinct roles in the evaluation pipeline.

The gold subset consists of 30 chat fragments selected to cover diverse discussion types. Each fragment is paired with a structured summary following the target schema (*Context, Decisions, Actions, Questions*).

Gold summaries are created using an LLM-assisted annotation pipeline based on GPT-4.1, followed by manual review and correction to ensure factual accuracy and structural consistency.

This subset is used exclusively for controlled diagnostic evaluation, enabling direct comparison of model behavior under identical conditions.
Due to the cost of manual validation, its size is intentionally limited; therefore, results are interpreted as comparative indicators rather than population-level performance estimates.

The remaining chat fragments are not manually annotated and are not used for computing reference-based evaluation metrics.

4.4 Data Annotation

Context

- The Context field contains a concise description of the main topic(s) discussed in the dialogue.
- It should summarize *what the conversation is about*, not how participants interact.
- Context must be inferable from the dialogue content and should avoid speculative or interpretive statements.
- Background chatter or off-topic messages should be ignored unless they affect the main discussion.

Decisions

- Decisions include explicit agreements, conclusions, or resolved choices made during the dialogue.
- A decision must reflect a shared or accepted outcome, not an individual opinion or suggestion.
- Tentative statements or unresolved proposals are excluded unless clearly confirmed.

- If no decisions are made, the field should be empty.

Actions

- Actions describe concrete tasks, next steps, or responsibilities arising from the dialogue.
- An action should be phrased as an executable item, even if the assignee or deadline is implicit.
- Requests, instructions, or commitments that imply future work are included.
- General intentions without a clear action component are excluded.

Questions

- Questions include explicit information-seeking or opinion-seeking interrogatives posed to the participants.
- Informal, conversational, or partially rhetorical questions are included if they invite a response.
- Clarification questions and requests for evaluation or feedback are included.
- Questions that are purely rhetorical or unrelated to the discussion topic are excluded.

5 Experiments

This section describes the evaluation methodology and experimental setup used to compare structured summaries generated by different language models.

5.1 Metrics

Evaluation is performed using a combination of component-wise automatic metrics and LLM-based judge scores.

For component-wise automatic evaluation, each field of the structured summary is assessed independently:

Context is evaluated using semantic similarity between generated and reference texts, computed as cosine similarity between sentence embeddings.

Decisions and Questions are evaluated using set-based overlap after text normalization, reporting precision, recall, and F1-score.

Actions are evaluated using slot-based matching, with the primary focus on the action description; assignee and deadline accuracy are reported as auxiliary metrics.

In addition to individual metrics, we report an aggregated automatic evaluation score (overall automatic score), computed as a weighted sum of key component-wise metrics:

$$overall\_auto = 0.35 \cdot ContextSim + 0.35 \cdot ActionsF1 + 0.15 \cdot DecisionsF1 + 0.15 \cdot QuestionsF1$$

This score serves as a diagnostic summary indicator to facilitate comparison across models.

It is not intended to represent a standardized benchmark metric and is used only for relative analysis within this study.

For Actions, Decisions, and Questions, micro-averaged F1 scores are used when computing the aggregated automatic score, as they better reflect extraction completeness under class imbalance.

In addition to automatic metrics, an LLM-based judge (gpt-4.1-mini) is used to assess overall summary quality in terms of usefulness, coherence, and faithfulness to the dialogue.

Judge-based evaluation is treated as a complementary signal and is not used as a replacement for reference-based automatic metrics.

5.2 Experiment Setup

All candidate summaries are generated using prompt-based inference without model fine-tuning. The same prompt and output schema are used across all models to ensure comparability.

Generation is performed on the gold subset for all models. Deterministic decoding settings are used to reduce output variance.

The gold subset is used exclusively for quantitative evaluation. All reported metrics are averaged across gold samples.

5.3 Baselines

The task of structured summarization of real-world work chats does not have established standard baselines or publicly reported benchmark results. Existing work on chat or meeting summarization typically produces unstructured free-text summaries, while action item and decision extraction approaches treat these components as separate information extraction tasks. As a result, direct comparison with prior systems is not feasible.

In this work, baseline systems are defined as general-purpose instruction-following language models with different capacities and deployment characteristics. All baseline models are evaluated under the same prompt-based generation setup and produce summaries in the same fixed structured format.

Three locally deployed models serve as lightweight and medium-capacity baselines:

Qwen2.5-3B-Instruct is used as a compact baseline representing low-resource local deployment scenarios.

LLaMA-3.1-8B-Instruct serves as a stronger local baseline with increased model capacity.

Mistral-7B-Instruct (local) serves as a medium-capacity baseline from the Mistral family. In combination with the API-based Mistral Large model, it supports a within-family comparison between local and API-based deployment settings, helping to disentangle the effects of deployment constraints and model scaling.

In addition, two API-based models are included as high-capacity baselines:

GPT-4o represents a state-of-the-art general-purpose language model with strong instruction-following and structured generation capabilities.

Mistral Large serves as a competitive non-GPT alternative, allowing comparison across different model families and training regimes, and supporting the within-family comparison with local Mistral-7B.

These baselines are chosen to reflect realistic deployment options rather than to exhaustively benchmark all available systems. The comparison focuses on how model capacity and deployment constraints affect the quality of structured summaries, rather than on achieving absolute state-of-the-art performance.

6 Results

This section presents the results of the experimental evaluation of the structured chat summarization approach.

All reference-based metrics are computed on the manually validated gold subset of 30 chat fragments (Section 4.3).

The results are interpreted as comparative and diagnostic, rather than population-level performance estimates.

6.1 Quantitative Evaluation

Table 1 reports the automatic evaluation metrics for all considered models. The comparison includes GPT-4o, Mistral Large, and several open-weight models deployed locally via Ollama.

Model	Context similarity	Actions F1 (micro)	Actions F1 (macro)	Decisions F1 (micro)	Questions F1 (micro)	Overall auto	Judge overall
openai_gpt4o	0.934	0.753	0.684	0.036	0.250	0.700	8.27
mistral_large	0.935	0.693	0.555	0.018	0.056	0.554	8.83
ollama_mistral_7b	0.876	0.519	0.455	0.000	0.023	0.517	7.80
ollama_llama3.1_8b	0.876	0.446	0.454	0.000	0.000	0.560	7.73
ollama_qwen2.5_3b	0.837	0.415	0.409	0.000	0.000	0.516	7.13

Table 1: Automatic and judge-based evaluation results (gold subset, n = 30).

Since no measurable differences were observed between action_thr = 0.75 and 0.80 across all metrics, results are reported for a single threshold (0.75) for clarity.

All models achieve high context similarity scores, indicating preservation of the main discussion topic.

However, substantial differences are observed in structured extraction quality.

GPT-4o achieves the highest overall automatic score, driven by strong performance on *Actions* extraction.

Mistral Large shows lower automatic scores but competitive judge-based ratings, suggesting more verbose but coherent summaries.

Open-weight local models demonstrate consistently lower extraction completeness, particularly for *Decisions* and *Questions*.

Varying the action filtering threshold (action_thr = 0.75 vs. 0.80) does not affect the results, indicating that the observed differences are driven by model generation behavior rather than post-processing.

6.2 Generation Behavior Analysis

To explain the observed metric differences, additional statistics were computed based on the number of generated actions relative to the gold reference.

Model	**Gold actions (avg)**	**Generated actions (avg)**	**Precision**	**Recall**	**Gen / Gold ratio**	**Behavioral pattern**
openai_gpt-4o	2.97	2.70	0.79	0.72	0.91	Conservative, decisions silent
mistral_large	2.97	3.00	0.69	0.70	1.01	Verbose, decisions and questions
ollama_mistral_7b	2.97	2.43	0.58	0.47	0.82	Questions verbose
ollama_llama3.1_8b	2.97	1.37	0.71	0.33	0.46	Actions and questions silent

ollama_qwen2.5_3b	2.97	1.37	0.66	0.30	0.46	Actions and decisions silent

Table 2: Action generation statistics and behavioral flags.

GPT-4o exhibits a conservative generation pattern, slightly under-generating actions while maintaining high precision.

Mistral Large produces a number of actions close to the gold reference, resulting in higher recall but increased verbosity.

Smaller models systematically under-generate structured elements, often omitting entire sections of the target schema.

These patterns explain the differences observed in F1 scores and highlight capacity-related limitations of smaller models.

6.3 Output Examples

Table 3 presents example outputs of the proposed summarization approach in Russian.

The examples illustrate typical model behavior, including accurate context extraction and variability in the completeness of structured fields. The examples are provided for illustration purposes and are not used for quantitative evaluation.

Dialogue fragment (shortene d)	**Generated structured summary gpt-4o**	**Generated structured summary ollama_mistral_7b**	**Gold structured summary**
2025-07-04 T11:50:35 | Speaker_A: У вас задание по интерфейсу? 2025-07-04 T11:50:42 | Speaker_E: есть api google gemini на 300$)) 2025-07-04 T11:50:43 | Speaker_C: по	{ "Context": "Обсуждается создание прототипа интерфейса для чатбота, включая визуализацию и интеграцию с ИИ.", "Decisions": [], "Actions": [{"assignee": "null", "description": "Оформить визуал бота и канала", "deadline": "null"}], "Questions": [	{ "Context": "Дискуссия о создании интерфейса для чатбота. Пользователь платит 15к долларов в год за просмотр документов внутри окна.", "Decisions": ["Необходимо прототипировать интерфейс для чатбота",	{ "Context": "Обсуждается создание прототипа продукта, в частности интерфейса для чатбота в Telegram и его визуального оформления. Участники уточняют, что подразумевается под интерфейсом и обсуждают задачи по интеграции ИИ с чатботом.",

прототипу продукта 2025-07-04 T11:50:51 | Speaker_A: Можете оформить визуал бота и канала	"Что мы хотим прототипировать?", "Какой интерфейс?"]}	"Использование API Google Gemini стоит 300$"], "Actions": [{"assignee": "Speaker_G", "description": "Показать как выглядят типичные сообщения", "deadline": null}, {"assignee": "Speaker_E", "description": "Оформить визуал бота и канала", "deadline": null}], "Questions": ["Что означает интерфейс?", "Какой интерфейс требуется?"]}}	"Decisions": ["Получен api google gemini на 300$.", "В качестве прототипа можно оформить визуал бота и канала и интегрировать ии в чатбот для телеграма."], "Actions": [], "Questions": [] }

Table 3: Output samples (Russian).

7 Conclusion

This work explores the task of structured summarization of real-world work chat conversations.

We propose a task-oriented summary format that explicitly separates conversational context, decisions, action items, and open questions, enabling fine-grained analysis of model behavior beyond plain abstractive summaries.

A dataset of anonymized work chat fragments is constructed, with a manually validated gold subset created via an LLM-assisted annotation pipeline followed by human correction.

Despite its limited size, the gold subset enables controlled diagnostic evaluation of structured extraction quality.

Experimental results demonstrate that different large language models exhibit distinct generation behaviors within the evaluated setup.

Larger proprietary models achieve higher precision and faithfulness, while larger open-weight models tend to generate more complete but verbose summaries.

Smaller open-weight models systematically under-generate structured elements, often omitting entire summary fields, indicating capacity-related limitations rather than failures of semantic understanding.

The combination of component-wise automatic metrics, an aggregated automatic score, and judge-based evaluation provides a practical framework for comparing structured summarization approaches under constrained annotation budgets.

Overall, the findings highlight key trade-offs between completeness, precision, and faithfulness in structured work-chat summarization and outline directions for future research on robust evaluation and schema-aware generation.

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